FLAVIO AVILA

Applied Scientist | Speech, Audio and Statistical Signal Processing, ML Evaluation, Production ML

Evanston, Illinois · flavio.rainho.avila@gmail.com · linkedin.com/in/flavioavila · flaviorainhoavila.github.io

SUMMARY

- Applied scientist and ML engineer with a doctorate in electrical engineering (Bayesian statistical signal processing), 15+ peer-reviewed publications across IEEE journals, Interspeech, ICASSP, WASPAA and EUSIPCO, and two US patent applications in voice-attribute machine learning.
- Seven years taking models from research into deployed systems at Verisk, Amazon and Agilent: speech and audio ML, detection and estimation on noisy real-world signals, fraud and risk detection, and the evaluation and regression practice that shows whether a model actually got better.
- Led a team of seven, including PhD scientists, after six years as a professor of signal processing, detection and estimation theory, and probability. US citizen, based in the Chicago area.

PROFESSIONAL INTERESTS

- Speech and audio ML, ASR robustness
- Detection and estimation on noisy signals
- ML evaluation and benchmarking
- Anomaly, fraud and risk detection
- ML for scientific instruments
- Research into production

PROFESSIONAL EXPERIENCE

Independent Research — Evanston, Illinois
ASR Robustness and Intelligibility-Aware Speech Enhancement — *2026 to present*

- Measuring how word error rate degrades as input quality drops, comparing a commercial recognizer with open-source models such as Whisper, and training enhancement models with learned quality and intelligibility predictors as objectives.

Agilent Technologies — Santa Clara, California
R&D AI/ML Engineer — *June 2023 to May 2026*

- Developed deep-learning detection and quantification models for noisy analytical-instrument signals and integrated them into commercial software under reliability and latency constraints, analyzing failure modes with the domain scientists who defined ground truth.
- Reformulated instrument method optimization as Bayesian inference, with the instrument response as a latent variable. Contributed deep-learning work on PFAS detection, presented at ASMS 2025.
- Part of the team that built the evaluation framework behind releases: benchmark suites, dataset versioning, MLflow experiment tracking and CI regression checks, so a change in performance could be attributed to the model rather than the data or the harness.

Amazon, Alexa — Sunnyvale, California
Applied Scientist — *June 2022 to January 2023*

- Built Transformer-based language models for question answering and product search serving consumer-scale live traffic, measured on impression rate, click-through and conversion.
- Designed evaluation datasets and human-in-the-loop annotation protocols for cases where automatic metrics could not establish whether a model had improved.

Verisk Financial/TransUnion and Verisk Analytics — Jersey City, New Jersey
Data Science Manager — *April 2022 to June 2022*
Manager of Analytics, AI/ML — *October 2020 to April 2022*
AI/ML Engineer and Fellow — *March 2019 to October 2020*

- Led a team of seven engineers and scientists that built a production speech analytics platform from an empty repository in Python, C and bash, for fraud and risk detection in life-insurance underwriting, deployed as an API across multiple commercial releases.
- Led two engineers on an in-house speech-recognition system that moved from Kaldi to wav2vec 2.0, reaching accuracy comparable to Azure, AWS and Google Cloud on our domain data.
- Built the signal pipeline, including diarization, normalization, PII suppression and a production quality gate for faulty and noisy audio; optimized diarization, the main latency bottleneck, to meet production SLAs.

- Defined the data strategy where labelled speech did not exist at scale: purchased corpora, commissioned annotation and automated label quality control, on HIPAA-regulated data. Results: two US patent applications and a peer-reviewed publication.

Rio de Janeiro State University Rio de Janeiro, Brazil
Associate Professor, Electronics and Telecommunications Engineering *March 2013 to March 2019*

- Led research in audio and speech signal processing, supervised graduate students, and published in IEEE journals and at Interspeech, ICASSP, WASPAA and EUSIPCO.
- Led a multidisciplinary research collaboration of physicists, mathematicians and engineers; taught signal processing, detection and estimation theory, and probability.

ADDITIONAL STRENGTHS

- Signal processing and statistics: spectral analysis, filtering, detection theory, Bayesian estimation and latent-variable models, Kalman filtering, hypothesis testing, bootstrap confidence intervals.
- Machine learning: deep learning for signals and sequences, Transformers, self-supervised speech models (wav2vec 2.0), Whisper, classification and regression, gradient boosting, anomaly detection.
- Speech: ASR development and evaluation, word error rate analysis, diarization, speech enhancement, speech quality and intelligibility, voice attributes.
- Python, PyTorch, NumPy, SciPy, scikit-learn, XGBoost, Hugging Face, SQL, FastAPI, MLflow, Docker, Git, CI/CD, AWS, MATLAB, C. Portuguese (native), English (fluent), Spanish (working).

EDUCATION

Doctor of Science, Electrical Engineering Federal University of Rio de Janeiro, 2012
Master of Science, Electrical Engineering Federal University of Rio de Janeiro
Bachelor of Science, Electrical Engineering Federal University of Rio de Janeiro

SELECTED PUBLICATIONS AND PATENTS

- 15+ peer-reviewed publications and a book chapter across IEEE journals, Interspeech, ICASSP, WASPAA and EUSIPCO, including:
- Bayesian restoration of audio signals degraded by impulsive noise modeled as individual pulses. IEEE Transactions on Audio, Speech, and Language Processing, 2012.
- A parametric objective quality assessment tool for speech signals degraded by acoustic echo. IEEE Transactions on Audio, Speech, and Language Processing, 2012.
- Investigating feature selection and explainability for COVID-19 diagnostics from cough sounds. Interspeech, 2021.
- Systems and Methods for Machine Learning of Voice Attributes. US Patent Applications 16/889,326 and 16/889,307, 2020.

DISTINCTIONS

- Google Scholar: 155 citations, h-index 7, i10-index 6.